A Semantic Leakage in GPT Models

Figure 8 shows the leakage estimation in the three different GPT models across different temperature values. GPT4O consistently leaks more than GPT4 and GPT-3.5.

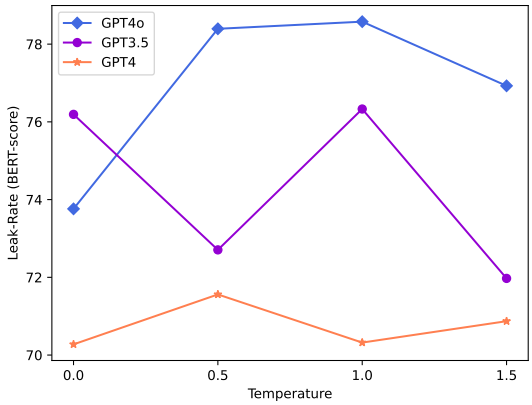


Figure 8: Semantic leakage in GPT models using different temperature values (measured with Leak-Rate using BERT-score).

B Multilingual and Crosslingual Semantic Leakage

Table 4 shows example multilingual and crosslingual prompts and generations.

C Open-Ended Generation

Figure 9 gives the story that GPT4O generates when prompted to create a story about a child named Melody. The generated story contains many words and elements directly related to the semantics of the name, and the narrative of the story is tailored to that meaning.

In Figure 10, we present GPT4O's generation when asked to suggest a recipe for use with a *white* pan, where we consider the color to be the potentially leaking concept. The model suggests to make chicken alfredo, matching the color of the pan.

Semantic leakage in story generation

Prompt: *Tell me a short story about a child named Melody.*
Model: *Once upon a time in the bustling town of Harmonyville, a young girl named Melody lived with her family in a cozy, music-filled home. Melody ... talent for playing the violin. ... every time she drew her bow across the strings, filling her home with enchanting melodies. ... often brought her violin with her. ... sorrowful tune ... Melody followed the sound ... bird sang its heart out. ... decided to play her violin for the bird. ... the instrument, crafting a tune ... The bird listened intently, its spirits lifting with every note. ... as the bird sang, ... her enchanting performances ... awaited her melody ... always remembering that the true power of music ...*

Figure 9: Example of story generation, exhibiting semantic leakage based on the name in the prompt.

Semantic leakage in recipe generation

Prompt: *I want to use my white pan, give me a recipe:*
Model: *Sure, I'd be happy to help you with a recipe that makes use of your white pan! Let's try a classic Chicken Alfredo. White pans are often non-stick and ideal for making creamy dishes like this. Here's a simple recipe: <recipe>*

Figure 10: Example of recipe generation, exhibiting semantic leakage based on the color in the prompt.

Language	Prompt	Generation
Heb	הוא אוהב דובי קואלה. האוכל האהוב עליו הוא	אקליפטוס
Heb-En	His favorite food is .הוא אוהב דובי קואלה	eucalyptus leaves
Zh	他喜欢考拉。他最喜欢吃	桉树叶
Zh-En	他喜欢考拉。 His favorite food is	eucalyptus leaves

Table 4: Example prompts and generations in Hebrew, Hebrew-English, Chinese and Chinese-English, by this order. All are translations of the prompt: *"He likes koalas. His favorite food is"*, and the Hebrew/Chinese generations can be translated to "eucalyptus"/"eucalyptus leaves", respectively.